Intrabeam Collisions, Gas and Electron Effects

USPAS - Beam Physics with Intense Space Charge

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Outline I

1. Coulomb collisions

2. Beam-gas scattering

3. Charge-changing processes

4. Gas pressure instability

5. Electron cloud processes

6. Electron-ion instability

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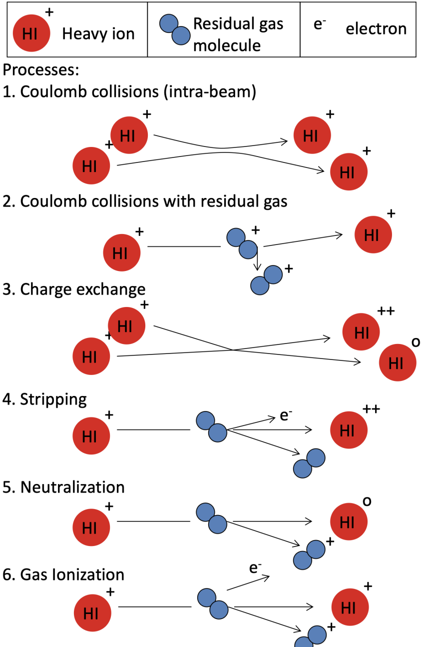
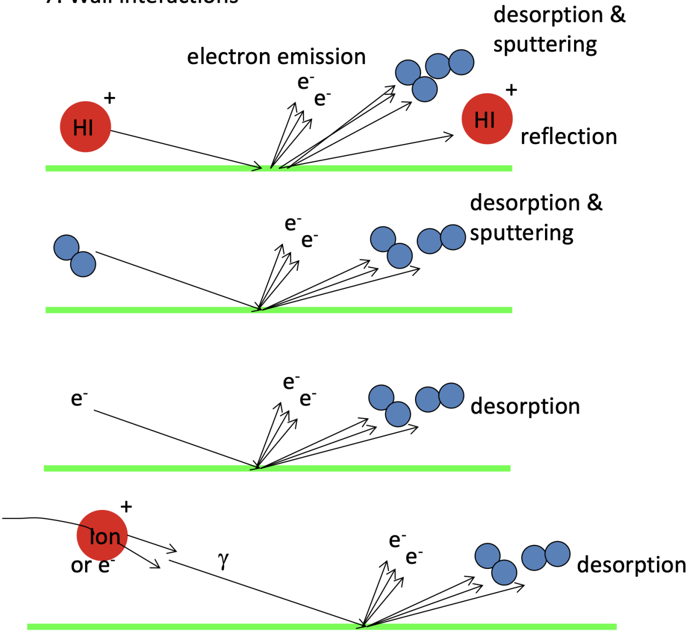
1.1 Gas and electron effects

- Effects depend strongly on particle mass and charge.
- Timescales are much different in circular accelerators vs. linacs ($t_{residence} \sim$ milliseconds to days vs. 10's of microseconds).
- Long vs. short pulse ($t_{pulse} \sim$ 10's of microseconds vs. 10's of nanoseconds).

1.2


HI
+
Heavy ion
Residual gas molecule
e⁻
electron
Processes:
1. Coulomb collisions (intra-beam)
HI
+
HI
+
HI
+
HI
+
2. Coulomb collisions with residual gas
HI
+
+
HI
+
+
3. Charge exchange
HI
+
HI
+
HI
++
HI
o
4. Stripping
HI
+
e⁻
HI
++
5. Neutralization
HI
+
HI
o
+
6. Gas Ionization
e⁻
HI
+
HI
+
+

7. Wall interactions



1.4 Coulomb collisions

Consider effects of Coulomb collisions in a continuous beam propagating through a continuous focusing channel with $T_\perp \neq T_\parallel$. (If $T_\perp \neq T_\parallel$, the beam is already relaxed.)

From Ichimaru and Rosenbluth [1]:

$$\frac{dT_\perp}{dt} = -\frac{1}{2}\frac{dT_\parallel}{dt} = \frac{T_\parallel - T_\perp}{\tau}.$$

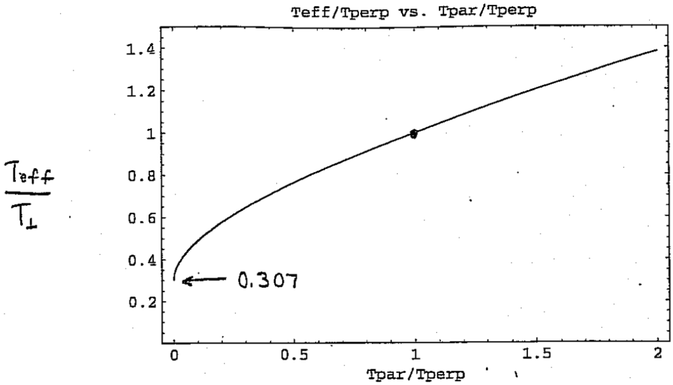
(Since $T_x = T_y = T_\perp$, $T_\parallel$ changes at twice the rate of $T_\perp$. Abd since $2k_BT_\perp + k_BT_\parallel = \text{constant}$.)

$$\begin{aligned}
\tau &= \text{relaxation time} \\
&= \frac{15(k_B\tilde{T}/mc^2)^{3/2}(4\pi\epsilon_0)^2 m^2 c^3}{8\pi^{1/2}q^4 \ln\Lambda\, n} \\
&= \left(\frac{15\pi^{1/2}}{8\ln\Lambda}\right)\frac{1}{\nu_c},
\end{aligned}$$

where ν_c is the collision frequency at thermal equilibrium:

$$\nu_c \sim \pi\left(\frac{q^2}{4\pi\epsilon_0 k_B T}\right)^2 n_0 \left(\frac{k_BT}{m}\right)^{1/2}.$$

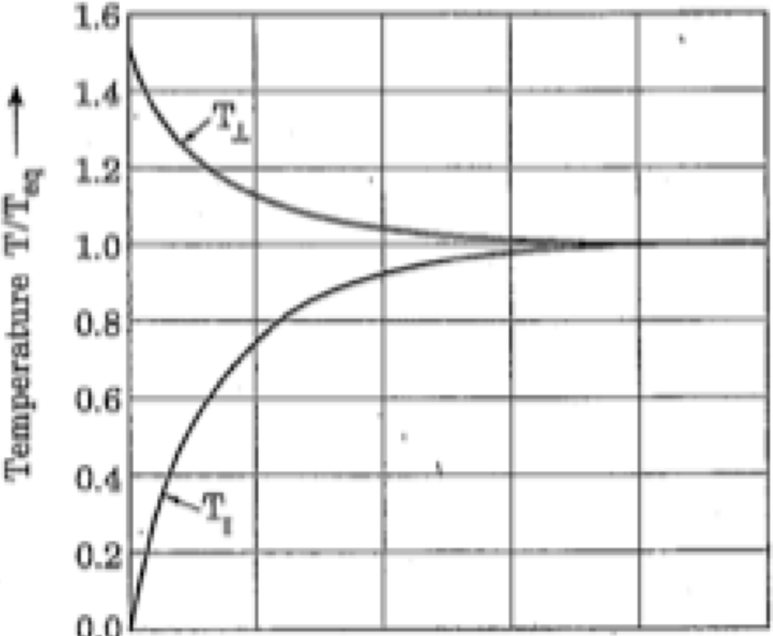
1.5



Teff/Tperp vs. Tpar/Tperp
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0.5
1
1.5
2
Tpar/Tperp
Teff
T⊥
0.307

[...]

1.6



1.7 The Boersch effect

Aren't collisions negligible? Putting in numbers for ions:

$$\tau_{\text{eff}} = 4.3 \cdot 10^{-4}[s] \left(\frac{A^{1/2}}{Z^4} \right) \left(\frac{k\tilde{T}}{[eV]} \right)^{3/2} \left(\frac{15}{\ln \Lambda} \right) \left(\frac{10^{10}[cm^{-3}]}{n} \right) \tag{1}$$

$$\Lambda = \frac{1.5 \cdot 10^5 (k_B T/[eV])^{3/2}}{Z^3 (n/10^{10}[cm^{-3}])} \tag{2}$$

Example: 2 MeV injector:

[...]

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2.1 Coulomb collisions in residual gas

Rate of change of momentum transverse to velocity:

$$\frac{dp_x}{dt} = \frac{ZZ_g e^2}{4\pi\epsilon_0 r^2}\frac{b}{r}$$

Total change in momentum:

$$\begin{aligned}
\Delta p &= \int_{-\infty}^{\infty} \frac{dp_x}{dt} dt \\
&= \int_{-\infty}^{\infty} \frac{dp_x}{dt}\frac{dt}{dz} dz \\
&= \frac{ZZ_g e^2 b}{4\pi\epsilon_0 v} \int_{\infty}^{\infty} \frac{dz}{(z^2+b^2)^{3/2}} \\
&= \frac{2ZZ_g e^2 b}{4\pi\epsilon_0 v b}.
\end{aligned}$$

Approximate deflection angle θ:

$$\theta \approx \frac{\Delta p}{p} = \frac{2ZZ_g e^2 b}{4\pi\epsilon_0 p v b}$$

This implies $db/d\theta \sim 1/\theta^2$

2.2 Multiple collisions

After traversing distance s and undergoing N_s collisions, the mean square angle $\langle \Theta^2 \rangle$ can be written

$$\begin{aligned}
&\langle \Theta^2 \rangle \\
&\quad = N_s \langle \theta \rangle^2 \\
&\quad = n_g \sigma_s s \langle \theta \rangle^2 \\
&\quad = 8\pi n_g \left(\frac{Z Z_g e^2}{4\pi\epsilon_0 m c^2 \gamma \beta^2} \right)^2 \ln\left(\frac{\theta_{max}}{\theta_{min}} \right) s
\end{aligned}$$

Writing

$$\langle \Theta^2 \rangle = \langle x'^2 \rangle + \langle y'^2 \rangle = 2\langle x'^2 \rangle,$$

and taking the s-derivative gives a dynamical equation for $\langle x'^2 \rangle$:

$$\frac{d}{ds}\langle x'^2 \rangle = 4\pi n_g \left(\frac{Z Z_g e^2}{4\pi\epsilon_0 m c^2 \gamma \beta^2} \right)^2 \ln\left(\frac{\theta_{max}}{\theta_{min}} \right) \tag{3}$$

$$\equiv C. \tag{4}$$

2.3 Jackson's approximation of θ_{min} and θ_{max}

Jackson argues that θ_{max} arises from the distributed nature of the nucleus and θ_{min} from the screening of electrons or the quantum uncertainty principle.

$$\ln\left(\frac{\theta_{max}}{\theta_{min}}\right) \approx 2\ln\left(204 Z_g^{-1/3}\right)$$

2.4 How does scattering affect the envelope equations?

Thin lens approximation: assume the scattering locally changes the transverse momentum without directly changing the position.

Let the initial phase space coordinates be $x = x_0$ and $x' = x_0'$. After an incremental distance δs,

$$x \to x_0 \tag{5}$$

$$x' \to x_0' + \delta x'. \tag{6}$$

We can then calculate the incremental change in the second-order moments:

$$delta\langle x'^2 \rangle = \langle ... \rangle$$

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3.1

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4.1

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5.1

Outline

6.1

6.2 References

[1] S. Ichimaru and M. N. Rosenbluth, *Relaxation Processes in Plasmas with Magnetic Field. Temperature Relaxations*, The Physics of Fluids **13**, 2778 (1970).